

Part 1

Problem 1

$$a: J = 3w_1^{(2)} - 6w_1 + w_2^{(2)} - 4w_2 + 8 \quad \frac{\partial J}{\partial w_1} = 6w_1 - 6$$

$$\frac{\partial J}{\partial w_2} = 2w_2 - 4$$

$$\nabla J(\vec{w}) = \begin{bmatrix} 6w_1 - 6 \\ 2w_2 - 4 \end{bmatrix}$$

$$b: \nabla J(\vec{w}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad 6w_1 - 6 = 0 \rightarrow w_1^* = 1 \quad 2w_2 - 4 = 0 \rightarrow w_2^* = 2$$

$$\vec{w}^* = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$c: \vec{w}^{(t+1)} = \vec{w}^{(t)} - \alpha \nabla J(\vec{w}^{(t)}) \quad \vec{w}^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\nabla J(0,0) = \begin{bmatrix} 6 \times 0 - 6 \\ 2 \times 0 - 4 \end{bmatrix} = \begin{bmatrix} -6 \\ -4 \end{bmatrix}$$

$$A (\alpha = 0,05) \bullet \vec{w}^{(1)} = \vec{w}^{(0)} - 0,05 \nabla J(\vec{w}^{(0)}) \Rightarrow \vec{w}^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0,05 \times \begin{bmatrix} -6 \\ -4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0,3 \\ 0,2 \end{bmatrix} = \begin{bmatrix} 0,3 \\ 0,2 \end{bmatrix} = \vec{w}_A^{(1)}$$

$$B (\alpha = 0,20) \bullet \vec{w}^{(1)} = \vec{w}^{(0)} - 0,2 \nabla J(\vec{w}^{(0)}) \Rightarrow \vec{w}^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0,2 \times \begin{bmatrix} -6 \\ -4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1,2 \\ 0,8 \end{bmatrix} = \begin{bmatrix} 1,2 \\ 0,8 \end{bmatrix} = \vec{w}_B^{(1)}$$

$$d: \vec{w}^* = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \|\vec{w}^{(1)} - \vec{w}^*\|_2 = \sqrt{(w_1^{(1)} - w_1^*)^2 + (w_2^{(1)} - w_2^*)^2}$$

$$A \rightarrow \vec{w}_A^{(1)} = \begin{bmatrix} 0,3 \\ 0,2 \end{bmatrix} \quad \|\vec{w}_A^{(1)} - \vec{w}^*\|_2 = \sqrt{(0,3 - 1)^2 + (0,2 - 2)^2}$$

$$= \sqrt{(-0,7)^2 + (-1,8)^2} = \sqrt{0,49 + 3,24} = \sqrt{3,73} \approx 1,9313$$

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$$B \rightarrow \| \vec{w}_B^{(1)} - \vec{w}^* \|_2 = \sqrt{(1,2 - 1)^2 + (0,8 - 2)^2} = \sqrt{(0,2)^2 + (-1,2)^2} \\ = \sqrt{0,04 + 1,44} = \sqrt{1,48} \approx 1,2166$$

B از نقطه شروع فاصله کمتری تا نقطه بهینه دارد پس در اولین update نرخ یادگیری 0,2 نسبت به 0,05 پیشرفت بیشتری به سمت نقطه بهینه ایجاد کرده است. دلیل این است که نسبت به $\vec{w}^{(t+1)} = \vec{w}^{(t)} - \alpha \nabla J$ با بزرگتر شدن α اندازه حرکت در جهت مخالف Gradient بزرگتر می شود.

چرا 0,2 بدون oscillation عمل می کند؟

تابع ما ^{convex} quadratic است و curvature آن در دو جهت برابر نیست. Hessian:

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$$H = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$$

برای Gradient Descent روی این تابع، شرط همگرا این: $0 < \alpha < \frac{2}{\lambda_{\max}}$ و $\lambda_{\max} = 6$

پس $0 < \alpha < \frac{2}{6} = 0,3333$ و $0,2$ و $0,05$ داخل محدوده هستند. پس با اندازه ای بزرگ هست که سریع تر حرکت کند ولی آنقدر بزرگ نیست که باعث نوسان یا divergence شود.

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Part 1

Problem 2

α : Linear Model Hypothesis: چون در Feature داریم،

$$\hat{y} = X\bar{w} + b \quad \text{یا} \quad \hat{y}_i = w_1 x_{i1} + w_2 x_{i2} + b \quad \leftarrow \text{مدل خطی}$$

$$\hat{\vec{y}} = X\bar{\vec{w}} + b \quad \text{پس} \quad \bar{\vec{w}} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}, \quad X = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} \quad \leftarrow \text{ماتریس ورودی}$$

MSE Cost Function:

Mean Squared Error $\rightarrow J(\bar{w}, b) = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)^2$

$$J = \frac{1}{3} [(\hat{y}_1 - y_1)^2 + (\hat{y}_2 - y_2)^2 + (\hat{y}_3 - y_3)^2] \quad \leftarrow \text{پس} \quad m = 3 \quad \leftarrow \text{تعداد نمونه ها}$$

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Gradient for Weights:

$$\nabla_{\bar{w}} J = \frac{2}{m} X^T (\hat{\vec{y}} - \vec{y}) \quad \leftarrow \text{فرد gradient نسبت به وزن ها}$$

$$\frac{\partial J}{\partial b} = \frac{2}{m} \sum_{i=1}^m (\hat{y}_i - y_i) \quad \leftarrow \text{برای bias}$$

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Gradient Descent Update:

$$\bar{w} \leftarrow \bar{w} - \alpha \nabla_{\bar{w}} J \quad \leftarrow \text{وزن ها}$$

$$b \leftarrow b - \alpha \frac{\partial J}{\partial b} \quad \leftarrow \text{Bias}$$

b: Epoch 1

$$w_1 = 0, w_2 = 0, b = 0 \quad \leftarrow \text{در ابتدا}$$

پس مدل اولی $\hat{y} = 0x_1 + 0x_2 + 0 \leftarrow$ یعنی prediction 0 می شود.

step 1 - Predictions $\rightarrow$ sample 1 $\rightarrow \hat{y}_1 = 0(1) + 0(0) + 0 \quad \hat{y}_1 = 0.0000$

sample 2 $\rightarrow \hat{y}_2 = 0(2) + 0(1) + 0 \quad \hat{y}_2 = 0.0000$ sample 3 $\rightarrow$

$\hat{y}_3 = 0(0) + 0(1) + 0 \quad \hat{y}_3 = 0.0000 \rightarrow$ پس $\hat{\vec{y}} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Step 2 - Residual Errors ->

$e_i = \hat{y}_i - y_i$ Residual
تعریف می کنیم

Sample 1 -> $e_1 = 0 - 2 = -2$ $e_1 = -2,000$ Sample 2 -> $e_2 = 0 - 5$

$= -5$ $e_2 = -5,0000$ Sample 3 -> $e_3 = 0 - 3 = -3$ $e_3 = -3,0000$

$$\vec{e} = \begin{bmatrix} -2 \\ -5 \\ -3 \end{bmatrix} \leftarrow \sigma_n$$

Step 3 - Gradient of w_1

$$\frac{\partial J}{\partial w_1} = \frac{2}{m} \sum_{i=1}^m e_i x_{i1} \rightarrow \frac{\partial J}{\partial w_1} = \frac{2}{3} [(-2)(1) + (-5)(2) + (-3)(0)]$$

$$= \frac{2}{3} [-2 - 10 + 0] = \frac{2}{3} (-12) \quad \frac{\partial J}{\partial w_1} = -8,0000$$

Step 4 - Gradient of w_2

$$\frac{\partial J}{\partial w_2} = \frac{2}{3} [(-2)(0) + (-5)(1) + (-3)(1)] = \frac{2}{3} [0 - 5 - 3]$$

$$= \frac{2}{3} (-8) \quad \frac{\partial J}{\partial w_2} = -5,3333$$

Step 5 - Gradient of Bias

$$\frac{\partial J}{\partial b} = \frac{2}{3} [-2 - 5 - 3] = \frac{2}{3} (-10)$$

$$\frac{\partial J}{\partial b} = -6,6667$$

Step 6 - Update Parameters

$$\alpha = 0,1$$

$$\text{update } w_1 \rightarrow w_1^{(1)} = w_1^0 - 0,1(-8) = 0 + 0,8 \quad w_1^{(1)} = 0,80000$$

$$\text{update } w_2 \rightarrow w_2^{(1)} = 0 - 0,1(-5,3333) \quad w_2^{(1)} = 0,5333$$

$$\text{update } b \rightarrow b^{(1)} = 0 - 0,1(-6,6667) \quad b^{(1)} = 0,6667$$

$$b^{(1)} = 0,6667, \quad \vec{w}^{(1)} = \begin{bmatrix} 0,8000 \\ 0,5333 \end{bmatrix} \leftarrow \text{Epoch 1 پایان}$$

sample	y	Prediction	Residual
1	2,0000	0,0000	-2,0000
2	5,0000	0,0000	-5,0000
3	3,0000	0,0000	-3,0000

Epoch 1

$$\text{Gradient} \rightarrow \nabla_w J = \begin{bmatrix} -8,0000 \\ -5,3333 \end{bmatrix}$$

$$\frac{\partial J}{\partial b} = -6,6667$$

$$\text{Parameters} \rightarrow w_1 = 0,8000, w_2 = 0,5333, b = 0,6667$$

$$\text{Epoch 2} \quad w_1 = 0,8 \quad w_2 = 0,5333 \quad b = 0,6667$$

$$\hat{y} = 0,8x_1 + 0,5333x_2 + 0,6667$$

$$\text{step 1 - Predictions} \rightarrow \text{Sample 1} \rightarrow x_1 = 1 \quad x_2 = 0$$

$$\hat{y}_1 = 0,8(1) + 0,5333(0) + 0,6667 = 0,8 + 0 + 0,6667 \quad \hat{y}_1 = 1,4667$$

$$\text{sample 2} \rightarrow x_1 = 2 \quad x_2 = 1 \quad \hat{y}_2 = 2,8000 \quad \text{sample 3} \rightarrow$$

$$x_1 = 0 \quad x_2 = 1 \quad \hat{y}_3 = 0,8(0) + 0,5333(1) + 0,6667 = 0 + 0,5333 + 0,6667 \quad \hat{y}_3 = 1,2000$$

$$\text{step 2 - Residuals} \quad e_i = \hat{y}_i - y_i \quad \text{sample 1} \rightarrow e_1 =$$

$$1,4667 - 2 \quad e_1 = -0,5333 \quad \text{sample 2} \rightarrow e_2 = 2,8000 - 5$$

$$e_2 = -2,2000 \quad \text{sample 3} \rightarrow e_3 = 1,2000 - 3 \quad e_3 = -1,8000$$

$$\vec{e} = \begin{bmatrix} -0,5333 \\ -2,2000 \\ -1,8000 \end{bmatrix}$$

$$\text{step 3 - Gradient } w_1$$

$$\frac{\partial J}{\partial w_1} = \frac{2}{3} [e_1 x_{11} + e_2 x_{21} + e_3 x_{31}] = \frac{2}{3} [(-0,5333)(1) +$$

$$(-2, 2)(2) + (-1, 8)(0)] = \frac{2}{3} [-0,5333 - 4,4 + 0] = \frac{2}{3} (-4,9333)$$

$$\frac{\delta J}{\delta w_1} = -3,2889$$

Step 4 - Gradient w_2

$$\frac{\delta J}{\delta w_2} = \frac{2}{3} [(-0,5333)(0) + (-2, 2)(1) + (-1, 8)(1)] =$$

$$\frac{2}{3} [-0,5333 - 2,2 - 1,8] = \frac{2}{3} (-4) \quad \frac{\delta J}{\delta w_2} = -2,6667$$

Step 5 - Gradient Bias

$$\frac{\delta J}{\delta b} = \frac{2}{3} [-0,5333 - 2,2 - 1,8] = \frac{2}{3} (-4,5333) \quad \frac{\delta J}{\delta b} = -3,0222$$

Step 6 - Update Parameters

$$w_1 \rightarrow w_1^{(2)} = 0,8 - 0,1(-3,2889) = 0,8 + 0,32889 \quad w_1^{(2)} = 1,1289$$

$$w_2 \rightarrow w_2^{(2)} = 0,5333 - 0,1(-2,6667) = 0,5333 + 0,26667 \quad w_2^{(2)} = 0,8000$$

$$b \rightarrow b^{(2)} = 0,6667 - 0,1(-3,0222) = 0,6667 + 0,3022 \quad b^{(2)} = 0,9689$$

$$b^{(2)} = 0,9689, \quad \vec{w}^{(2)} = \begin{bmatrix} 1,1289 \\ 0,8000 \end{bmatrix} \leftarrow \text{Epoch 2}$$

sample	y	Prediction	Residual
1	2,0000	1,4667	-0,5333
2	5,0000	2,8000	-2,2000
3	3,0000	1,2000	-1,8000

Epoch 2

$$\text{Gradient} \rightarrow \nabla_w J = \begin{bmatrix} -3,2889 \\ -2,6667 \end{bmatrix}$$

$$\frac{\delta J}{\delta b} = -3,0222$$

$$\text{Parameters} \rightarrow w_1 = 1,1289, \quad w_2 = 0,8000, \quad b = 0,9689$$